RESEARCH REPORT · VITKI

Fully tested finding

The runestones were raised by the water of their time – 64 per cent within 500 metres of the water of AD 1050, against 46 per cent of the land around them

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Timeline of the study: The analysis plan was locked 23 October 2026 · A test run of the script with no results kept 23 October 2026 · The distances were computed 23 October 2026 · The addendum for moved stones was locked 31 October 2026 · The statistics were run and read 31 October 2026 · The addendum plan for the sensitivity analyses was locked and run 1 November 2026

ABSTRACT

Background and aim. Where the runestones were raised has long been debated: by grave fields, by roads and bridges, by farms and boundaries. Water is harder to measure, because the land has risen since the Viking Age and today's shore is not the stones' shore. The question here is whether the stones were raised closer to the water of their own time than the land around them, and whether that association remains when nearness to the grave fields is held equal.

Material. 1,099 Viking-Age runestones not counted as moved in the locked corpus (253 of them lost and measured at their recorded place), in Uppland, Södermanland, Östergötland, Småland, Västergötland and on Öland, and 10,990 random points on land within five kilometres of the stones. The water of AD 1050 from the shore displacement model of the Geological Survey of Sweden (SGU), the grave fields from the Swedish National Heritage Board's register of ancient monuments.

Method. Pre-registered study. The distance to the nearest water is compared between stones and points with Wilcoxon's rank-sum test and the AUC, the probability that a stone stands closer than a point. Internal replication in two halves, seven sensitivity tests in the plan and nine sensitivity analyses under a locked addendum plan.

Results. 64 per cent of the stones stand within 500 metres of the water of AD 1050, against 46 per cent of the points; the median distance is 305 against 555 metres, AUC 0.63, $p < 0.0001$. It holds in both halves, in five of six provinces, for other years between 550 and 1550, with an independent elevation model and in every sensitivity analysis, including against the land in the same parish. The grave fields draw more

strongly: 67 against 24 per cent within 500 metres, AUC 0.78. Both associations remain when the other is held equal. Of the 657 stones that stand in place according to three sources, 63 per cent stand within 500 metres of the water (AUC 0.629), and without the 253 lost stones and the 43 that the corrected rule for moved stones removes, the AUC is 0.627 for the water and 0.794 for the grave fields. The water of the time fits the stones slightly better than today's, but the difference is small.
Conclusion. Those who raised the runestones chose places near the water, and even more often near the grave fields, within districts that already lay by water. The main weakness is that the model of the water of AD 1050 is coarse: 108 of the stones fall in the model's water.

Keywords: *runestones, Viking Age, placement, shore displacement, land uplift, grave fields, spatial statistics, pre-registration*

HIGHLIGHTS

- Those who raised the runestones chose places near the water of their time, not only because the district lay by the water but also within the district.
- The grave fields drew even more strongly, and each association remains when the other is held equal: the places most likely to have a runestone lay near both the water and the grave fields.
- Together with the text being turned towards the road, it points to stones raised to be seen by those who travelled.
- The weakness is the model of the water of AD 1050, which is coarse near the shores; the next test is a finer elevation model.

1. Introduction

Where the runestones were raised has long been debated. Gunnar Ekholm (1950, pp. 138–143) objected to the view of the stones as monuments of the grave fields: in Uppland, he argued, they largely followed the old routes by land and water, and a stone on Ängby land near Knivsta was raised to be seen from the river, "the route of travel in ancient times" (*forntidens samfärdsled*). Mats G. Larsson (1990) found in a survey of the stones' placement that graves, roads with bridges and fords, and probable property boundaries were about equally common settings (Larsson 2010, p. 249). Torun Zachrisson (1998) has linked the stones to farm, boundary and grave field, and Lydia Klos (2009) has argued that most were raised at grave fields. Mats G. Larsson (2010, p. 252) has objected that Klos thereby overstates the grave fields at the expense of other settings, above all the property boundaries (see also McLeod 2014). Per Stille (2014, p. 142) found in Tiohärad in Småland that 20 of the 53 stones whose place can be determined with more or less certainty stood by water, at watercourses or bridge sites, where a road presumably also ran. The bridge stones, which tell of bridges and causeways over water and wet ground, show that water belonged to the stones' setting.

Measuring the nearness to water is hard. The land has risen since the Viking Age, in the Mälaren valley by about five metres, so a stone that stood by a bay may now stand on dry land, and a stone by today's shore may have stood far from the water. Klos (2009) also measured the distance to water, but as a rule without the sea level of the Viking Age, which Larsson (2010, p. 251) saw as the basic flaw of that part of her study: Lake Mälaren became an inland lake instead of the inlet of the sea

that it was. SGU's shore displacement model (SGU 2022, after Påsse & Daniels 2015) gives an open model for the whole country of where the water lay at different times.

We know of no study that has measured the distance to the water of the stones' own time for the whole Swedish material, with a model of shore displacement, without the stones counted as moved, and with a comparison with the land in the same district. The aim is to do so and test whether the association remains when nearness to the grave fields is held equal.

2. Pre-registration: hypothesis and plan

The hypotheses, measures and rules of interpretation were written down in an analysis plan on 23 October 2026, before any distance had been computed. The plan has the SHA-256 checksum `4d-db7d6cecb470a1…` and was committed to version control the same evening. Before the lock, only the data sources had been fetched, the layers listed, the number of stones per province counted and the mid-years of the style groups compiled to choose the year (median 1035). A test run the same evening, after the plan had its checksum but before the commit, ran through the tests but printed only its progress and the numbers of stones, random points and grave fields, no test results; no results from it survive, and the statistics were first read on 31 October.

The direction was expected. The project's earlier measure, with a coarse land uplift and with moved stones, had already shown that the stones stand closer to water and grave fields than the land around them. The study therefore tests whether a known pattern holds with better data, not a new one.

The plan's hypotheses, verbatim (translated): **H1 (primary): the stones stand closer to the water of AD 1050 than the random points. H2: the nearness is clearer relative to the water of the time than to today's. H3: the grave fields, on the same selection** (the stones stand closer to registered grave fields than the random points). H1 *holds* if five conditions are met: p below 0.05 and AUC above 0.5 overall, the same in both halves, the same when stones within 200 metres of each other are merged into one place, the same for four other years within 200 years (750, 950, 1150 and 1250), and the same with only the sea, without the model's lakes. H3 holds if it holds overall, in both halves and for the places. H2 is called clearer only if the whole 95 % interval for the difference lies above zero.

Two addenda were locked later, both before they were computed. On 31 October (SHA-256 `dc63b72b3147f9c2…`), before the statistics were run: H1 and H3 are also tested on the stones that stand in place according to three sources, because the rule for moved stones had been corrected on 27 October, after the plan was locked. On 1 November (SHA-256 `2993273eff5006c6…`), when the plan's results were known: nine fixed sensitivity analyses for distances, corresponding to those the project makes for other findings. They are confirmatory in the sense that the steps and limits were locked before they were computed, but they were chosen after the main result was known.

3. Material

- **The stones.** Swedish Viking-Age runestones in the Scandinavian Runic-text Database (Rundata), in the version locked in the plan. Of them we took those that have coordinates and are not counted as moved under the rule as it stood when the plan was locked: Rundata's placement does not mention a church, church porch, wall, museum, collection, park or similar place to which stones have been taken, and the site is not a church, churchyard, church ruin, vicarage, monastery, castle, school or museum. The rule was corrected on 27 October 2026, after the plan was locked, so that it also counts stones whose placement says that they have been moved or names another place than the find-spot; under the corrected rule 43 of the stones here are moved. Lost stones are not removed by the rule: 253 of the stones are lost and measured at their recorded place. So not all stones in the selection still stand in place; the sensitivity analyses in 5.3 test the result without them. Only provinces with at least 20 stones in the selection are tested: Uppland (623), Södermanland (234), Östergötland (89), Småland (69), Västergötland (64) and Öland (20), 1,099 in all. The 45 stones in the other provinces are not tested. Runestones only.
- **The water.** SGU's shore displacement model, open data version 1.0 (2022, CC BY 4.0). It is built on shore displacement curves across the country, a model of land uplift and an elevation model with a resolution of 50 metres. The water of AD 1050 is the model's sea and lakes in the slice 900 years before 1950; today's water is the slice for 1949. The distances are computed on a grid of 25-metre cells.
- **The grave fields.** The Swedish National Heritage Board's register of ancient monuments via K-samsök (open data, CC0), retrieved on 12 October 2026: 18,633 registered grave fields with coordinates.
- **The independent elevation model.** Copernicus DEM GLO-30 (about 30 metres), with the province's land uplift since the Viking Age as the height limit.
- **The dating** by the stone's ornamental style (Gräslund 2006): 587 of the stones.
- **The halves** are the project's fixed division parish by parish: 543 and 556 stones.

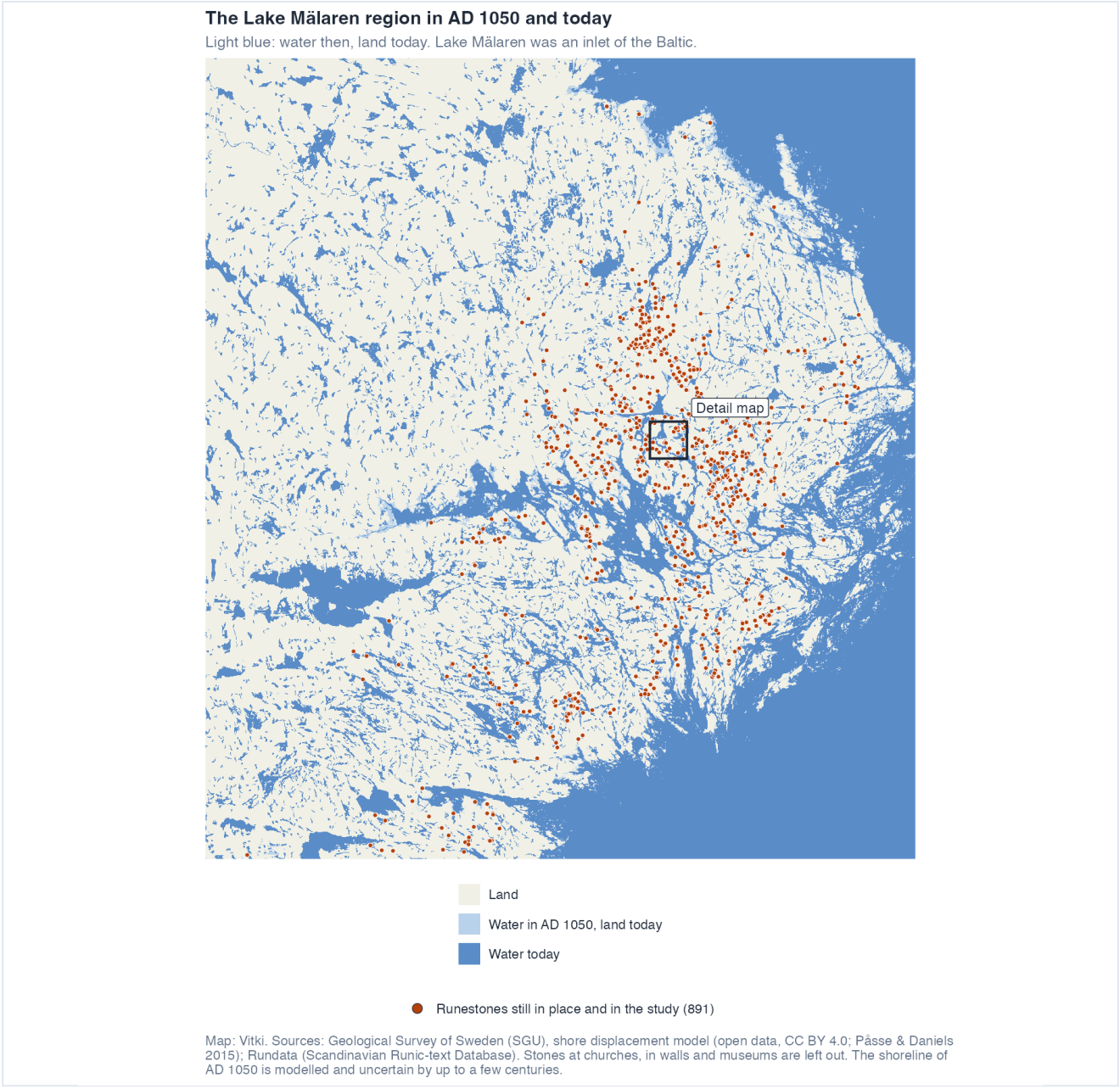


Figure 1. The Lake Mälaren region in AD 1050 and today. Light blue is water then but land today. The box shows the area of figure 2. The points are the 891 stones of the study that lie within the map. The legend in the image calls them stones still in place, which is not true of all: 205 of them are lost and drawn at their recorded place, and 36 count as moved under the corrected rule of 27 October 2026, one of them both (240 stones in all). Map: Vitki, after SGU's model (CC BY 4.0) and Rundata.

4. Method

4.1 The measure and the comparison. For every stone and every point the distance is computed to the nearest water of AD 1050, to today's water and to the nearest grave field; a point in the water gets the distance zero. The comparison is a null model: for each province ten random points per stone are drawn, uniformly within five kilometres of the province's stones, and only points that are

land both in AD 1050 and in 1949 are kept. The points thus lie in the districts where stones were raised, so the test concerns where in the district the stone stands, not where the districts lie. Stones and points are compared with Wilcoxon's rank-sum test (Wilcoxon 1945; Mann & Whitney 1947), one-sided. The effect is given as the medians, the share within 500 metres and the AUC, the probability that a randomly chosen stone stands closer than a randomly chosen point (0.5 means no difference). Figure 2 shows the measurement in one district.

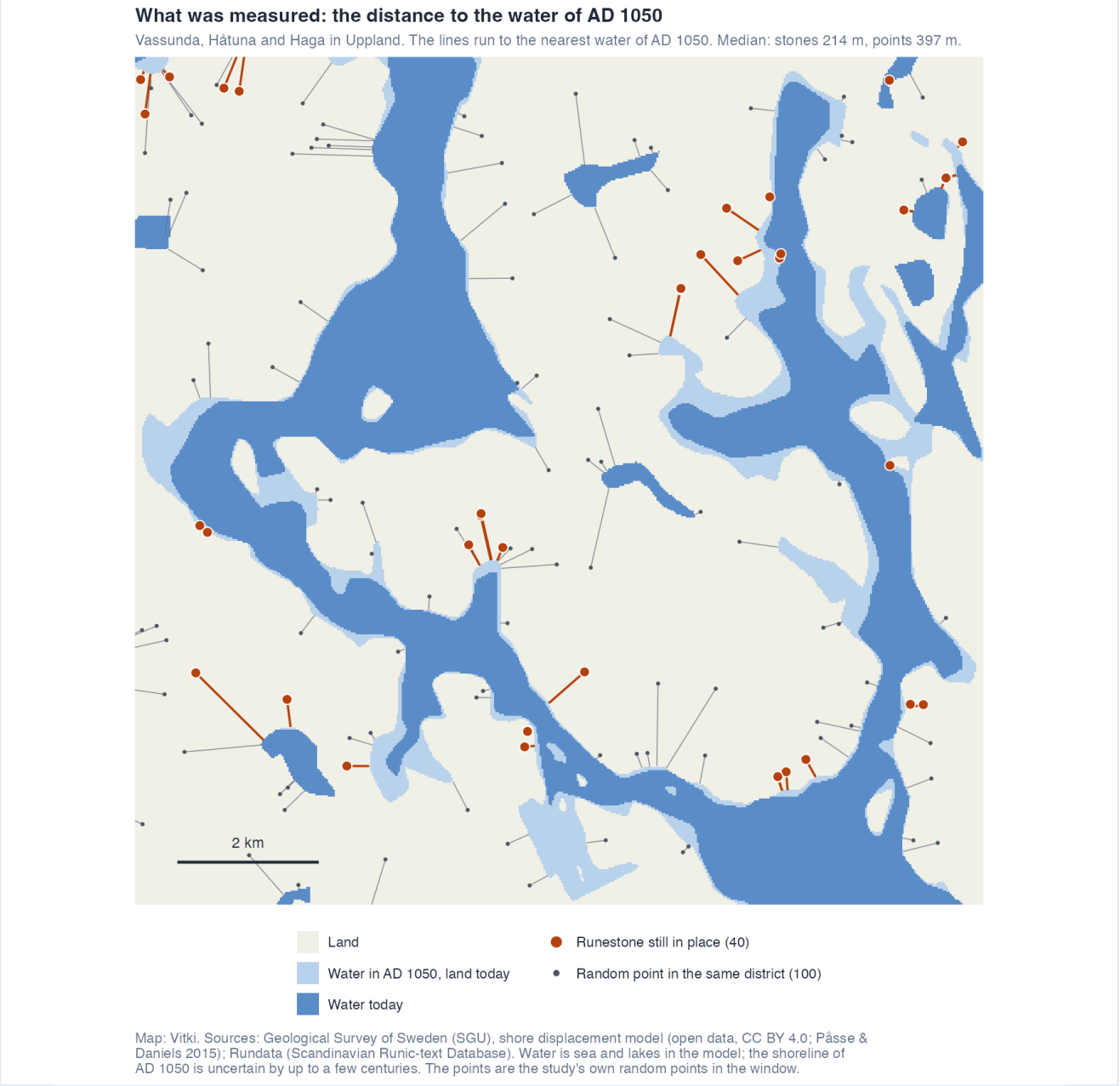


Figure 2. What was measured, around Vassunda, Håtuna and Haga in Uppland: the distance from each stone (red) and random point (grey) to the nearest water of AD 1050. Median here: stones 214 metres, points 397. The legend in the image calls all 40 stones still in place; 5 of them are lost and drawn at their recorded place, and 2 count as moved under the corrected rule. Map: Vitki, after SGU's model and Rundata.

4.2 Correction and replication. Each hypothesis is tested in each half on its own. The provinces are tested separately with Benjamini–Hochberg correction (Benjamini & Hochberg 1995) over the six,

and the secondary tests likewise over the five. The difference between the water of the time and today's (H2) gets a 95 % interval from 2,000 bootstrap samples in which the stones are drawn place by place.

4.3 The sensitivity analyses. The plan's seven tests and its descriptive comparison with moved stones, the addendum's test and the nine steps of the addendum plan are given with their results in 5.2 and 5.3. The parish as a random effect is a logistic regression (Bates et al. 2015) in which the points get the parish of the nearest stone. When stones are removed, the points that belong to them are removed too. A step holds when the AUC is above 0.5 and p below 0.05; without one province at a time, significance is required only where at least two thirds of the stones remain.

4.4 Deviations from the plan. The statistics were run eight days after the distances were computed, after the addendum for moved stones; the plan's tests are unchanged. The 15,590 distances that reached the edge of the computation tiles, nearly all for the sea alone far inland, were recomputed on a grid of 100-metre cells over the whole province; the plan gives no cell size. The sensitivity analyses of the addendum plan were added after the plan's results were known (see 2). Otherwise none.

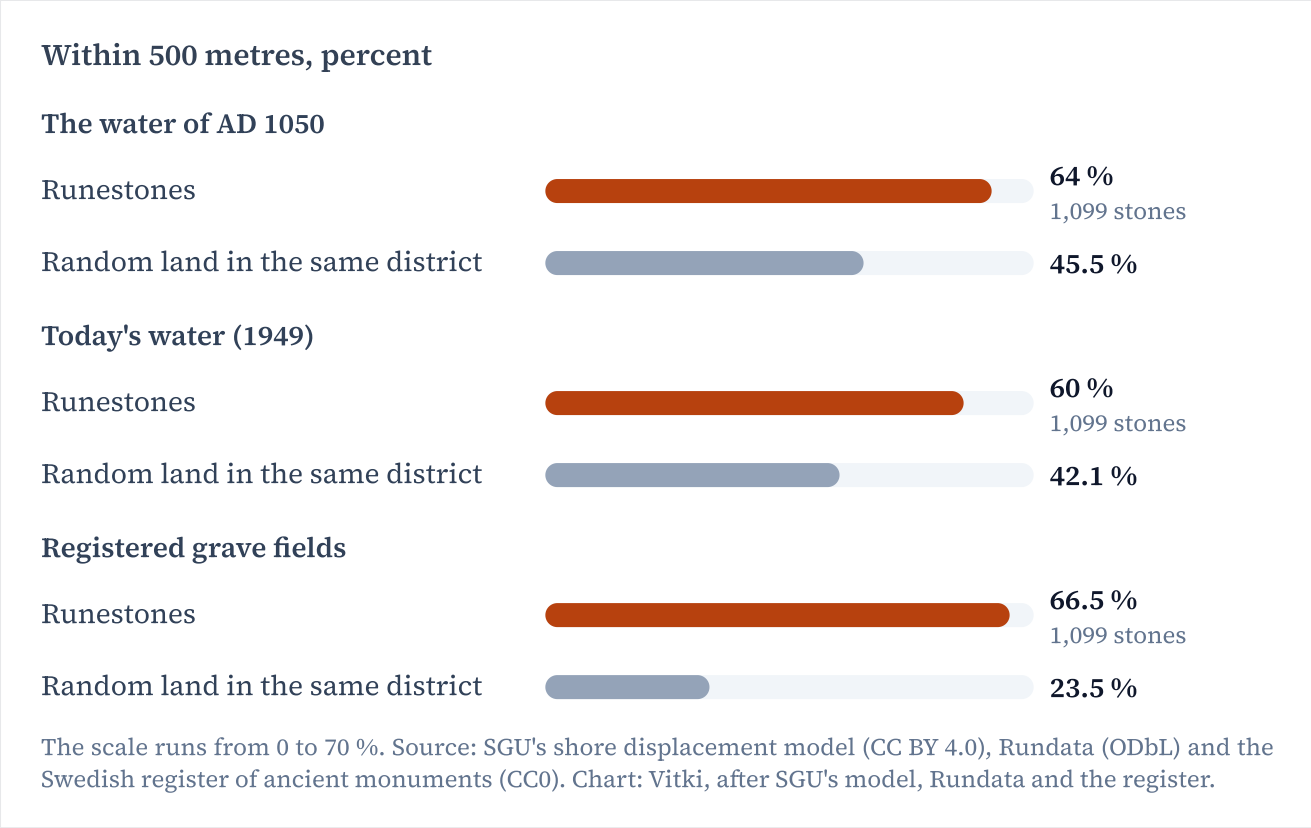
5. Results

5.1 The water of AD 1050 (H1), the grave fields (H3) and today's water (H2). Of the 1,099 stones, 64 per cent stand within 500 metres of the water of AD 1050, against 46 per cent of the 10,990 points; within one kilometre 83 against 76 per cent. The median distance is 305 against 555 metres, AUC 0.629, p < 0.0001. Half A (543 stones in 190 parishes): AUC 0.639, p < 0.0001. Half B (556 stones in 187 parishes): AUC 0.619, p < 0.0001. With 935 places instead of stones: AUC 0.626, p < 0.0001. H1 *holds* on all five conditions. In plain words: compare a runestone with a randomly chosen place in the same district, and the runestone stands closer to the water of its time in 63 cases out of 100; almost two in three stones stand within 500 metres of the water, against fewer than half of the land.

By province, with q corrected over the six: AUC 0.63 in Uppland, 0.67 in Södermanland, 0.59 in Östergötland, 0.59 in Småland and 0.61 in Västergötland, all q < 0.01, and 0.61 on Öland, q = 0.057, where the 20 stones go the same way but are not enough.

The grave fields (H3): 67 per cent of the stones stand within 500 metres of a registered grave field, against 24 per cent of the points; the median is 304 against 1,045 metres, AUC 0.784, p < 0.0001. Half A: AUC 0.790, half B: AUC 0.778, the places: AUC 0.771, all p < 0.0001. H3 *holds*, in the same five provinces as H1. In plain words: two in three stones stand within 500 metres of a grave field, against one in four places in the district.

Today's water (H2): relative to the water of 1949, 60 per cent of the stones stand within 500 metres, against 42 per cent of the points, AUC 0.620. The difference in AUC from the water of the time is 0.009, 95 % interval 0.002 to 0.017. The whole interval lies above zero, so by the plan the nearness is *clearer* relative to the water of the time, but the difference is small. It shows in Uppland, Södermanland, Östergötland and on Öland; in Småland and Västergötland today's water fits as well or slightly better.



5.2 The plan's sensitivity tests. All hold for H1:

- Other years: AUC between 0.62 and 0.64 for all six, from AD 550 to AD 1550, all $p < 0.0001$.
- Only the sea, without the model's lakes: 1,099 stones, median 2,975 against 4,496 m, AUC 0.57, $p < 0.0001$. In Östergötland, Småland and Västergötland the test says nothing, since most stones and points there lie tens of kilometres from the sea.
- Without the 108 stones that stand in the model's water of AD 1050: 991 stones, median 372 against 555 m, AUC 0.59, $p < 0.0001$.
- The independent elevation model: land uplift one metre lower: median 335 against 570 m, AUC 0.63, $p < 0.0001$; equal: median 301 against 560 m, AUC 0.64, $p < 0.0001$; one metre higher: median 285 against 552 m, AUC 0.65, $p < 0.0001$. The pattern thus does not depend on SGU's elevation model.
- Points within two kilometres: AUC 0.628; within ten kilometres: AUC 0.627. For the grave fields the difference grows with the radius (AUC 0.707 and 0.829), while the radius changes almost nothing for the water.

The plan's descriptive comparison with the moved stones included (2,167 stones, 1,068 of them moved, and new random points around them): AUC 0.650 against the water of AD 1050. The moved stones on their own, measured where they stand today, mostly at churches: median 285 metres, AUC 0.659. They thus stand even closer to the water of the time than the stones in the selection, so counting them would have made the association somewhat stronger, not weaker.

Water and grave fields together, in a logistic regression with the province held constant and the distances on a logarithmic scale, taken as the distance plus 250 metres so that a distance of zero can also be used: when the distance to the water plus 250 metres halves, the odds that the place has

a runestone become 1.72 times as large (95 % interval 1.58–1.87), with the distance to the grave fields equal; for the grave fields the odds ratio is 3.46 (3.18–3.76). Both hold in both halves and after correction. In plain words: the water is not merely a sign that the grave fields lie there, nor the other way round; the grave fields weigh about twice as much.

The other secondary tests do not hold: the share within one kilometre of the water is the same for early and late stones (84 per cent), Christian markers make no difference ($p = 0.47$), and the bridge and road stones stand somewhat closer (207 against 309 metres in median) but not reliably ($p = 0.059$, $q = 0.10$).

5.3 The sensitivity analyses. Only stones that stand in place according to three sources (the locked corpus, the corrected rule for moved and lost stones and Runor 2020; 657 stones): 63 per cent within 500 metres of the water of AD 1050, against 46 per cent of the points, AUC 0.629; grave fields AUC 0.786, both $p < 0.0001$. In the nine analyses of the addendum plan every step holds for both the water and the grave fields. For the water:

- The halves: see 5.1.
- Without one province at a time: AUC between 0.62 and 0.63, also without Uppland, when only 43 per cent of the stones remain ($p < 0.0001$).
- The parish as the unit (373 parishes): 373 stones, median 336 against 555 m, AUC 0.61, $p < 0.0001$.
- The parish as a random effect: odds ratio 1.82 when the distance plus 250 metres halves (95 % interval 1.68–1.98), $p < 0.0001$; the model converged without warnings. In plain words: also compared with the land in the same parish, the stones stand closer to the water.
- One stone per certain carver (67 carvers): 767 stones, median 313 against 570 m, AUC 0.62, $p < 0.0001$.
- Without re-erected stones (101 removed): 998 stones, median 307 against 559 m, AUC 0.63, $p < 0.0001$.
- Without stones that Runor gives as moved (177 removed): 922 stones, median 305 against 553 m, AUC 0.63, $p < 0.0001$.
- Only stones still standing under the corrected rule of 27 October 2026 (295 removed: 253 lost and 43 moved, one of them both): 804 stones, median 305 against 562 m, AUC 0.63, $p < 0.0001$; grave fields AUC 0.794.
- The early stones, before AD 1050: 335 stones, median 276 against 575 m, AUC 0.65, $p < 0.0001$. The late: 252 stones, median 336 against 534 m, AUC 0.62, $p < 0.0001$.

For the grave fields the AUC lies between 0.72 and 0.80 in every step, without one province at a time between 0.74 and 0.79, and the parish as a random effect gives the odds ratio 3.57 (3.27–3.89), $p < 0.0001$.

The study stands alone, so the correction applies within the plan and not among the associations the project adjusts jointly. No hand-coded traits are involved; the source of error is the model of the water (see 6).

6. Discussion

What the finding says. Those who raised the runestones chose places near the water. The comparison concerns the land in the same district, and the association remains parish by parish, so it is not only because the districts lay by water. Nor does it depend on a few provinces, places or carvers, on stones that have been moved, lost or re-erected, or on SGU's elevation model. It held both before and after AD 1050. The grave fields draw more strongly, and each association remains when the other is held equal: the most likely places for a runestone lay near both water and grave fields.

The water of the time and today's. That the water of the time fits only slightly better than today's does not mean that the land uplift is unimportant. In most districts the shore has moved less than the hundreds of metres that separate the stones from the points (figure 2), and the lakes remain. The model for AD 1050 is needed above all so that the finding is not an artefact of today's coast.

Against earlier research. That two in three stones stand within 500 metres of a grave field supports Klos (2009) and agrees with stones in family groups standing at grave fields more often than single stones (Kvant 2026a). That the water remains when the grave fields are held equal at the same time bears out Larsson's (2010) and McLeod's (2014) point that the grave fields are not the whole answer. The finding fits with the stones' text being turned mostly towards the road (Kvant 2026b): the stones were raised where people travelled, by land and by water, as Ekholm (1950) argued from single stones in Uppland. That is an interpretation; the study shows where the stones stand, not why.

Weaknesses, in order of importance.

1. **The model of the water is coarse.** SGU's model is regional, with an elevation model of 50 metres, time errors of up to 500 years and approximate lakes; rivers, streams and wetlands are missing. 108 stones fall in the model's water of AD 1050, where no stone can have been raised. Without them the association is weaker but clear (AUC 0.59), and the independent elevation model gives the same result.
2. **The direction was expected.** The project's earlier measure showed the same with coarser data. The study confirms a known pattern; it discovers nothing new.
3. **The sensitivity analyses were chosen after the main result.** The steps and limits were locked before they were computed, but after the plan's results were known.
4. **The stones' places.** The main selection rests on the rule for moved stones as it stood before the correction of 27 October 2026, and it includes 253 lost stones, which are measured at their recorded place. Without them and the 43 stones that the corrected rule counts as moved, the result is the same (AUC 0.627 for the water and 0.794 for the grave fields). A stone may also have been moved a little without it being recorded. Three sources for moved stones and a word list for re-erected stones give the same result, but do not catch everything.
5. **The grave fields** are unevenly surveyed in different districts and mostly undated; some may be younger or older than the stones.
6. **Öland** has too few stones to show anything for certain.

What would overturn the finding. If the association with water disappeared with a finer elevation model, for example Lantmäteriet's at one metre's resolution, and shore displacement curves for

each district, or if the stones' original places on the oldest maps lay systematically farther from the water.

7. Conclusion

The runestones were raised by the water of their time: almost two in three stand within 500 metres of the water as it lay in AD 1050, against fewer than half of the land around them, and the association withstands every sensitivity analysis. The grave fields drew even more strongly, and each association remains when the other is held equal. The next step is a finer elevation model and the oldest maps.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se), with the changes made after that version up to 1 November 2026. When report version 2 was written there was no release of Vitki with a DOI, so the code state is given as a commit: the analysis, the result files and the maps are those of commit `4f80a63` (2026-11-01). All code, the analysis plans and the result files are openly available on GitHub under GPL-3.0.

- The analysis plans: `analysplaner/2026-10-23-strand-och-gravfalt.md`, `analysplaner/2026-10-31-strand-tillagg.md` and `analysplaner/2026-11-01-strand-trappan.md`, with their checksums in the `.sha256` files beside them.
- The corpus: fingerprint `d6cffd1578418f74` (locked in the plan). That folder is no longer in the cache, so the path in `utdata/strand/config.R` points to a missing file; the same CSV survives as `data/cache/r/corpus/5fdfd1e095bfb6fb/corpus.csv`, with the SHA-256 that the plan gives, `4fca4baf47867e5c294d4c97d7dd34496d7fcc07996f68efddec1261a2bcb5ba`. Anyone rerunning the analysis points the path there. The corrected rule for moved and lost stones comes from `e83202e01a5e6d27`. The corpus files are built locally from Rundata and are not in the repository.
- The check for report version 2: `utdata/strand/kontroll_v2.py` counts the lost and moved stones in the selection and in the maps and reads the log of the test run; the result is in `utdata/strand/kontroll_v2.json`. The file and the builder of version 2 are in commit 6eae0af (2026-11-05).
- The source passages for report version 3: `utdata/strand/kallcitat_v3.json` holds the verbatim quotations from Stille (2014), Ekholm (1950) and McLeod (2014) that the pages and numbers in the introduction and the references are checked against.
- The analysis, one step per R process: `utdata/strand/01_avstand.R`, `01b_marginal.R`, `02_r6.R`, `03_statistik.R`, `04_figurer.R` and the sensitivity analyses `05_trappa.R`. Results in `utdata/strand/resultat.json` and `trappan.json`, the stones in `stenar.csv`, the full account in `resultat.md` (Swedish).
- SGU's model (`strandforskjutning_1_900.gpkg`, `strandforskjutning_1000_1900.gpkg`) and Copernicus DEM are downloaded to a cache and are not in the repository; their checksums are in the plan.

- The maps: `utdata/strand/06_illustration.R` (the window in figure 2 was chosen as the 12-kilometre window in the Mälaren region with the most stones times the share of land that was water in AD 1050 but is land today). The maps' legends say "still in place" of all points; the maps have not been redrawn, and the captions say what the points are.
- The report is built by `utdata/blogg/build/build_rapport_strand.py`, which reads every number from the result files and stops if anything disagrees.

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Change log

Version 1, 2026-11-01. Published 2026-11-01. The same morning, before publication, the draft was changed (commit 4f80a63, whose message mentions only the maps): the maps got the source line "Map: Vitki, after ...", the sentence of aim in the introduction was shortened, and the sentence on the grave fields in the plan's test with other radii for the random points was removed from 5.2 without a reason being recorded. The published version therefore lacked the grave fields' result in that test, and the plan's comparison with the moved stones had never been included; the plan says that both are to be reported.

Version 2, 2026-11-05. Corrections after a review before archiving in Zenodo. The analysis, the result files and the conclusion are unchanged; the maps have not been redrawn.

- **The selection.** Version 1 described the 1,099 stones as stones that still stand in place. That was wrong for 295 of them: 253 are lost and measured at their recorded place, and 43 count as moved under the rule that was corrected on 27 October 2026, after the plan was locked (one stone is both). The abstract, the introduction, the description of the material, the captions of figures 1 and 2, the step with the corrected rule in 5.3 and weakness 4 now say what the selection is, and the Material section describes the rule that was used. The conclusion stands: without the 295 stones the AUC is 0.627 for the water and 0.794 for the grave fields, and of the 657 stones that stand in place according to three sources, 63 per cent stand within 500 metres of the water of AD 1050.
- **The test run of 23 October** ran through the tests, which version 1 did not say; it printed only its progress. The timeline now has the day the distances were computed.
- **Two results required by the plan** were missing: the grave fields with random points within two and ten kilometres, and the comparison with the moved stones. They are now in 5.2.
- **Earlier research.** The objection to Klos is now attributed to Mats G. Larsson (2010), whom McLeod (2014) cites, and the introduction names earlier work on the stones and the water (Ekholm 1950, Larsson 1990, Stille 2014) and Larsson's criticism of how Klos measured the water.
- **Water and grave fields.** Version 1 called the associations independent, which says more than the model shows; the text now says that each remains when the other is held equal, and that the odds ratios are per halving of the distance plus 250 metres.
- **Sources and form.** The Copernicus attribution follows the wording of the licence, the register's licence (CC0) is given, the uncited reference to the project's method description is removed (the two other Kvant 2026 are now a and b), the English links go to the English pages, the chart says

that Vitki made it, and Data and code gives the code state as a commit and says that the locked corpus's folder is missing.

Version 3, 2026-11-05. A second review before archiving in Zenodo checked what version 2 changed. The analysis, the result files and the maps are unchanged.

- **Sources.** McLeod (2014) has pages 1–8, not 1–18. Ekholm's (1950) objection to the grave fields begins on p. 138, so the pages are now given as 138–143. Stille (2014) is now given with what p. 142 says: 20 of the 53 stones whose place can be determined stood by water; version 2 wrote "a large share". The article and the reference have been checked against the journal.
- **The test run of 23 October** printed its progress and the numbers of stones, random points and grave fields, but no test results; version 2 said that it printed no figures. The timeline now has the test run.
- **Figure 1.** The caption says that one of the stones is both lost and moved, so that the 205 and the 36 make 240.
- **The essentials box** no longer says that a stone was most often raised where both the water and the grave fields were, which the report does not measure; it says what the model shows, that the places most likely to have a runestone lay near both.
- **Changes that version 2 made but did not log:** the pages of Ekholm (1950, pp. 137–147) and McLeod (2014) were given, the licence of Rundata (ODbL) was added to the chart's source line, and the discussion added that the finding fits what Ekholm (1950) argued from single stones in Uppland.
- **Data and code** gives the commit of the check file and builder of version 2 (6eae0af) and lists the file with the quoted source passages.

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Cite as

Kvant, V. (2026). The runestones were raised by the water of their time – 64 per cent within 500 metres of the water of AD 1050, against 46 per cent of the land around them. Research report, report version 3; analysis in Vitki 3.0.0. https://vitkirunor.se/en/forskningsrapporter/runstenarna-vid-sin-tids-vatten